

Class 9: Halloween Candy Mini-Project

Shuk Yee Leung (PID: A18050465)

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Today we will take a wee step back to some data we can taste and explore the correlation structure and principal components of some Halloween candy.

Data Import

```
candy_file <- "candy-data.csv"
candy = read.csv(candy_file, row.names=1)
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0

	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294

One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

Q1. How many different candy types are in this dataset?

```
nrow(candy)
```

```
[1] 85
```

Q2. How many fruity candy types are in the dataset?

```
sum(candy$fruity)
```

```
[1] 38
```

1. What is your favorite candy?

Q3. What is your favorite candy in the dataset and what is its winpercent value?

```
candy["Twix", ]$winpercent
```

```
[1] 81.64291
```

```
candy["Skittles", ]
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer	hard	bar
NA	NA	NA	NA	NA	NA		NA	NA
pluribus	sugarpercent	pricepercent	winpercent					
NA	NA	NA	NA	NA				

```
View(candy)
```

Q4. What is the winpercent value for “Kit Kat”?

```
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

2. Exploratory Analysis

We can use the **skimr** package to get a quick overview of a given dataset. This can be useful for the first time you encounter a new dataset.

```
skimr::skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

It looks like the last column `candy$winpercent` is on a different scale to all others.

Q7. What do you think a zero and one represent for the `candy$chocolate` column?

```
skimr::skim(candy$chocolate)
```

Table 3: Data summary

Name	candy\$chocolate
Number of rows	85
Number of columns	1
Column type frequency:	
numeric	1
Group variables	None

Variable type: numeric

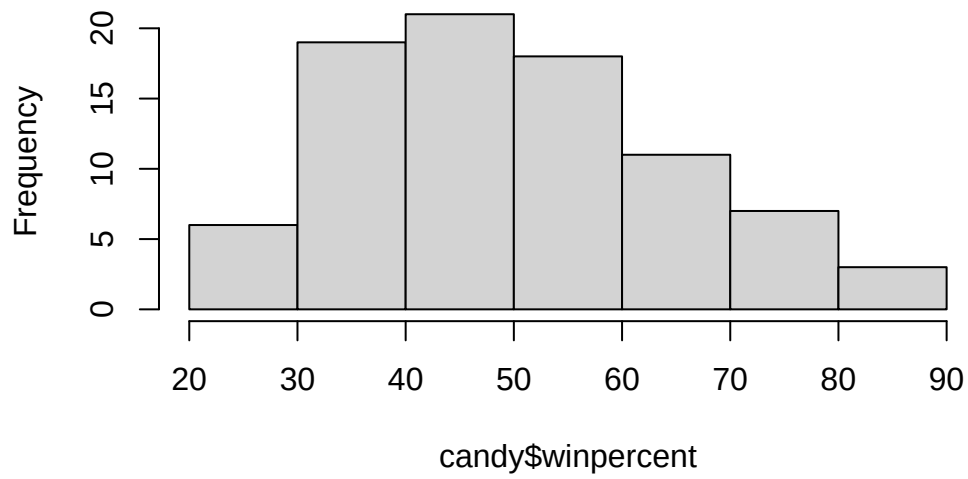
skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
data	0	1	0.44	0.5	0	0	0	1	1	

zero means the candy is not chocolate, one means the candy is chocolate.

Q8. Plot a histogram of winpercent values

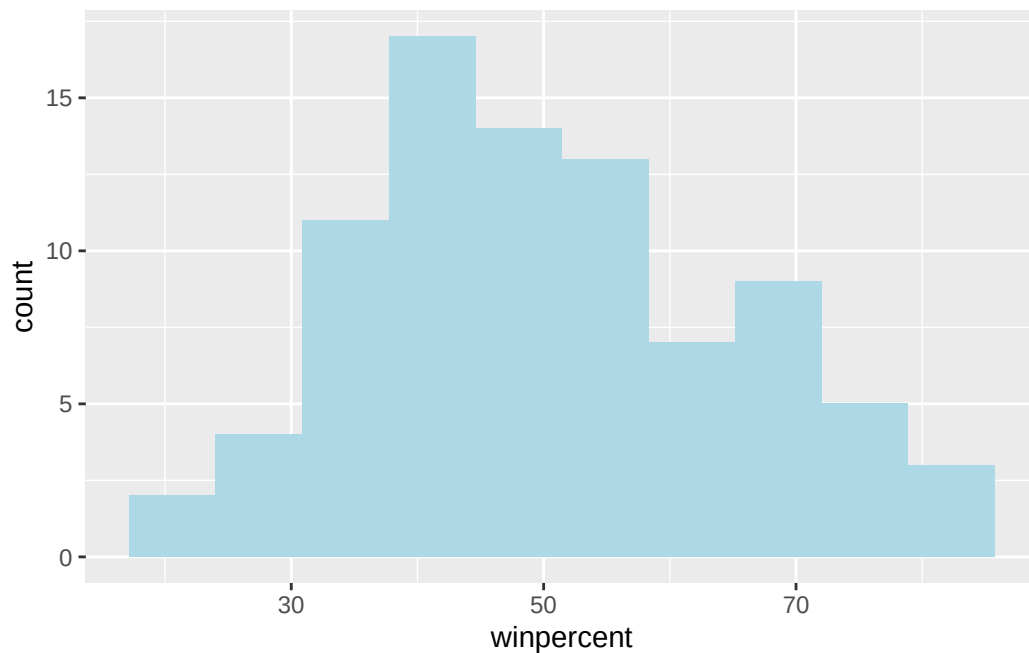
```
hist(candy$winpercent)
```

Histogram of candy\$winpercent



```
library(ggplot2)

ggplot(candy) +
  aes(winpercent) +
  geom_histogram(bins=10, fill="lightblue")
```



Q9. Is the distribution of winpercent values symmetrical?

No.

Q10. Is the center of the distribution above or below 50%?

```
summary(candy$winpercent)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
22.45	39.14	47.83	50.32	59.86	84.18

Median is below 50%.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
choc.inds <- candy$chocolate == 1
choc.candy <- candy[choc.inds, ]
choc.win <- choc.candy$winpercent
mean(choc.win)
```

```
[1] 60.92153
```

```
fruit.win <- candy[ as.logical(candy$fruity), ]$winpercent  
mean(fruit.win)
```

```
[1] 44.11974
```

On average, chocolate candy is higher ranked than fruit candy.

Q12. Is this difference statistically significant?

```
ans <- t.test(choc.win, fruit.win)  
ans
```

Welch Two Sample t-test

```
data:  choc.win and fruit.win  
t = 6.2582, df = 68.882, p-value = 2.871e-08  
alternative hypothesis: true difference in means is not equal to 0  
95 percent confidence interval:  
 11.44563 22.15795  
sample estimates:  
mean of x mean of y  
 60.92153  44.11974
```

Yes, there is a statistically different with a p-value of 2.8713778×10^{-8} .

```
ans$p.value
```

```
[1] 2.871378e-08
```

3. Overall Canday Rankings

Q13. What are the five least liked candy types in this set?

There are two related functions that can help here, one is the classic `sort()` and `order()`.

```
x <- c(5,10,1,4)  
sort(x, decreasing = T)
```

```
[1] 10 5 4 1
```

```
order(x)
```

```
[1] 3 4 1 2
```

```
inds <- order(candy$winpercent)
head( candy[inds, ], 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat		
Nik L Nip	0	1	0		0	0		
Boston Baked Beans	0	0	0		1	0		
Chiclets	0	1	0		0	0		
Super Bubble	0	1	0		0	0		
Jawbusters	0	1	0		0	0		
	crisped	rice	wafer	hard	bar	pluribus	sugar	percent
Nik L Nip		0	0	0		1	0.197	0.976
Boston Baked Beans		0	0	0		1	0.313	0.511
Chiclets		0	0	0		1	0.046	0.325
Super Bubble		0	0	0		0	0.162	0.116
Jawbusters		0	1	0		1	0.093	0.511
	winpercent							
Nik L Nip	22.44534							
Boston Baked Beans	23.41782							
Chiclets	24.52499							
Super Bubble	27.30386							
Jawbusters	28.12744							

Q14. What are the top 5 all time favorite candy types out of this set?

```
tail( candy[inds, ], 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat		
Snickers	1	0	1		1	1		
Kit Kat	1	0	0		0	0		
Twix	1	0	1		0	0		
Reese's Miniatures	1	0	0		1	0		
Reese's Peanut Butter cup	1	0	0		1	0		
	crisped	rice	wafer	hard	bar	pluribus	sugar	percent
Snickers		0	0	1		0		0.546

Kit Kat	1	0	1	0	0.313
Twix	1	0	1	0	0.546
Reese's Miniatures	0	0	0	0	0.034
Reese's Peanut Butter cup	0	0	0	0	0.720

	pricepercent	winpercent
Snickers	0.651	76.67378
Kit Kat	0.511	76.76860
Twix	0.906	81.64291
Reese's Miniatures	0.279	81.86626
Reese's Peanut Butter cup	0.651	84.18029

```
inds <- order(candy$winpercent, decreasing = T)
head( candy[inds, ], 5)
```

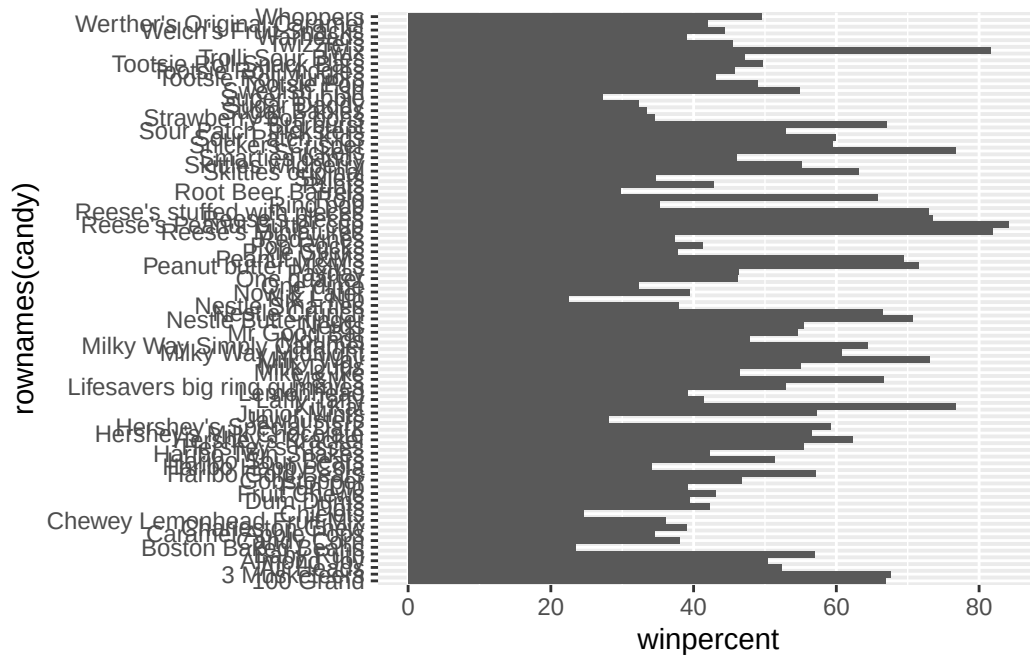
	chocolate	fruity	caramel	peanut	almond	nougat
Reese's Peanut Butter cup	1	0	0		1	0
Reese's Miniatures	1	0	0		1	0
Twix	1	0	1		0	0
Kit Kat	1	0	0		0	0
Snickers	1	0	1		1	1

	crisped	rice	wafer	hard	bar	pluribus	sugar
Reese's Peanut Butter cup		0	0	0		0	0.720
Reese's Miniatures		0	0	0		0	0.034
Twix		1	0	1		0	0.546
Kit Kat		1	0	1		0	0.313
Snickers		0	0	1		0	0.546

	pricepercent	winpercent
Reese's Peanut Butter cup	0.651	84.18029
Reese's Miniatures	0.279	81.86626
Twix	0.906	81.64291
Kit Kat	0.511	76.76860
Snickers	0.651	76.67378

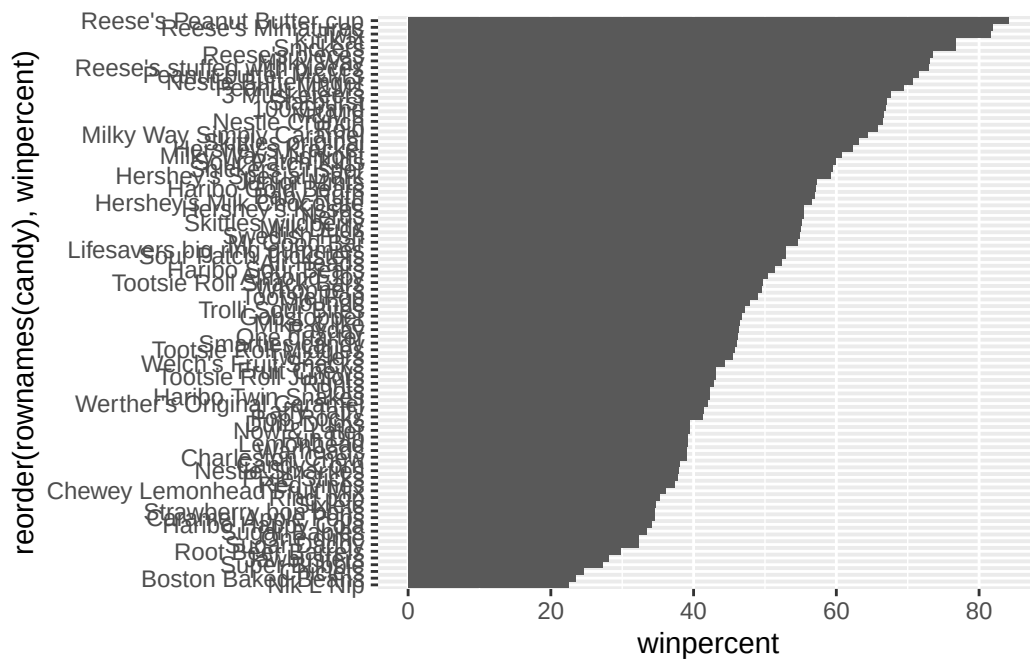
Q15. Make a first barplot of candy ranking based on winpercent values. Make a bar plot with ggplot and order it by winpercent values

```
ggplot (candy) +
  aes(winpercent, rownames(candy)) +
  geom_col()
```

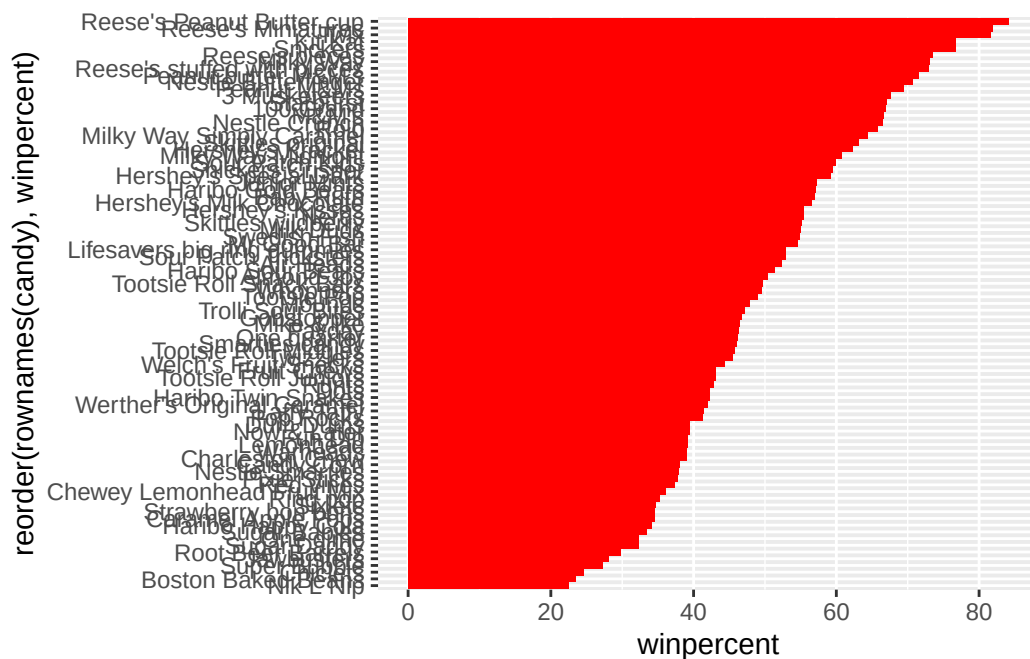


. Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent?

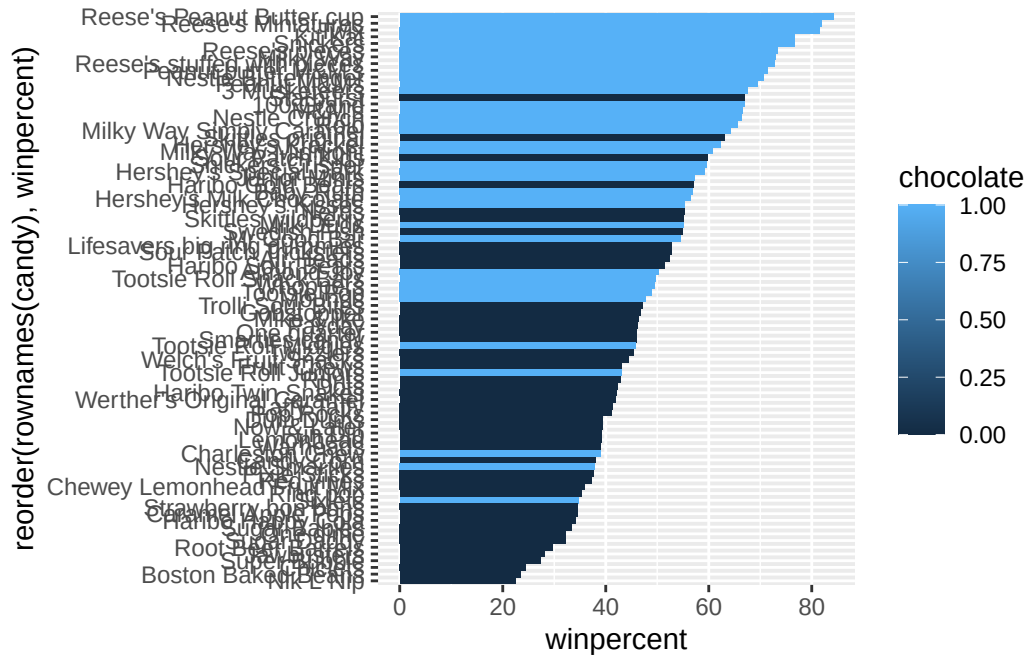
```
ggplot (candy) +  
  aes(winpercent, reorder(rownames(candy),winpercent)) +  
  geom_col()
```



```
ggplot (candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill="red")
```



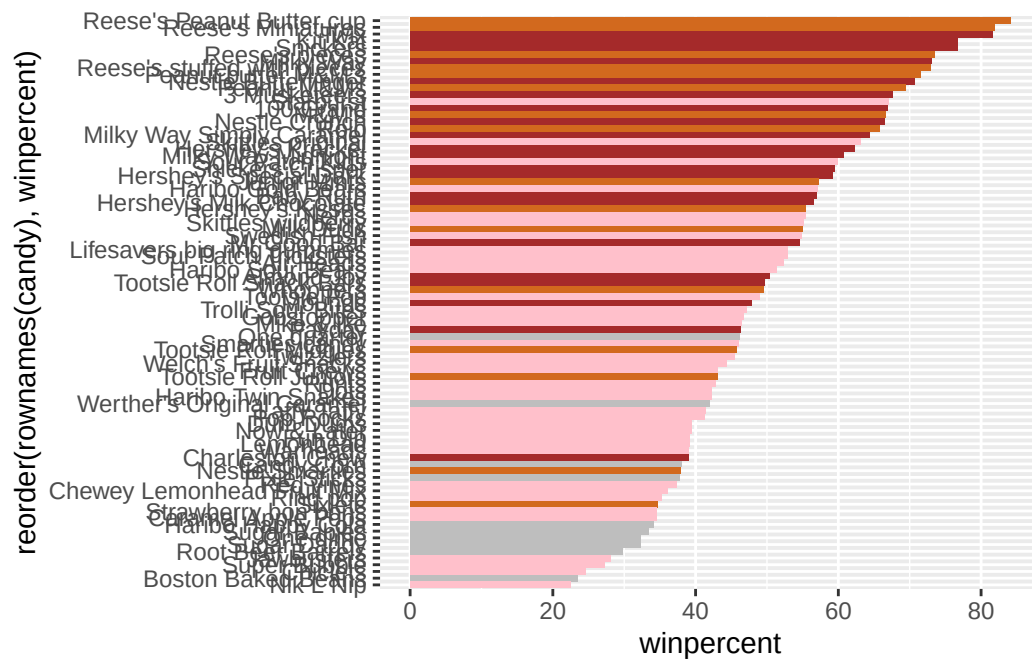
```
ggplot (candy) +
  aes(x= winpercent, y=reorder(rownames(candy),winpercent), fill=chocolate) +
  geom_col()
```



Here we want a custom color vector to color each bar the way we want with `chocolate` and `fruity` candy together with whether it is `bar` or not

```
mycols <- rep("gray", nrow(candy))
mycols[as.logical(candy$chocolate)] <- "chocolate"
mycols[as.logical(candy$fruity)] <- "pink"
mycols[as.logical(candy$bar)] <- "brown"

#mycols
ggplot (candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=mycols)
```



```
ggsave("mybarplot.png", width=3, height=8)
```

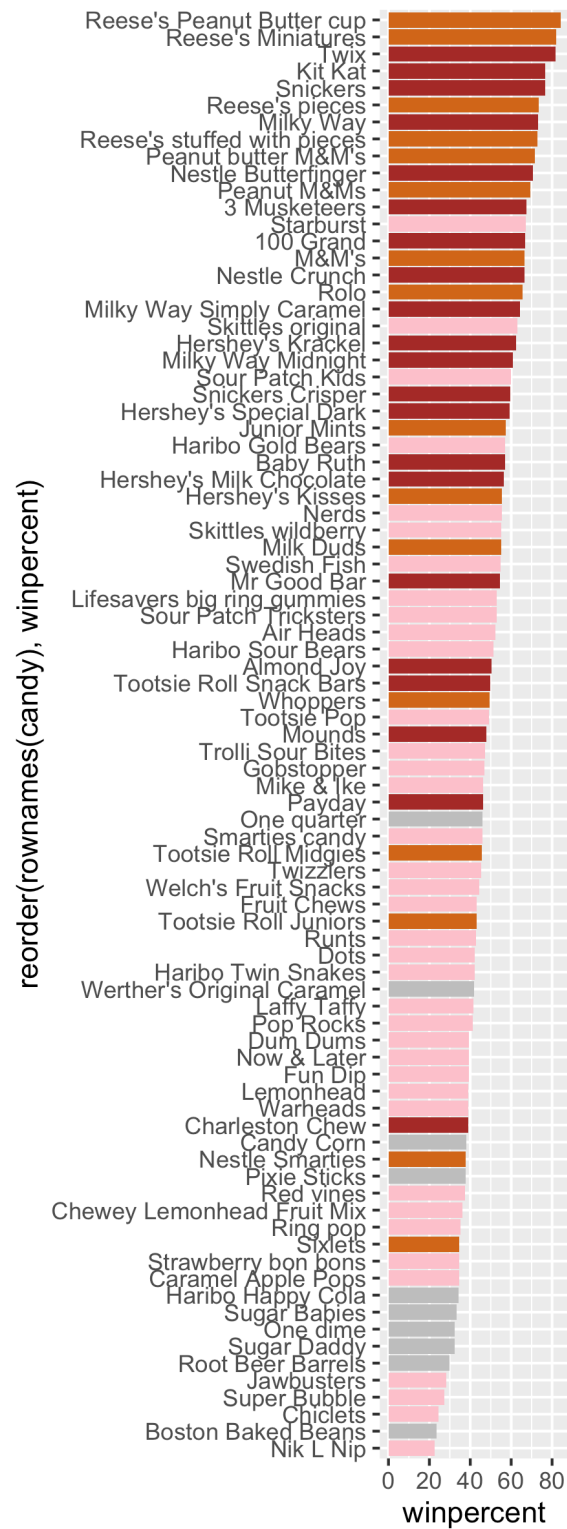


Figure 1: My silly barplot image

Q17. What is the worst ranked chocolate candy?

Sixlets

Q18. What is the best ranked fruity candy?

Starburst

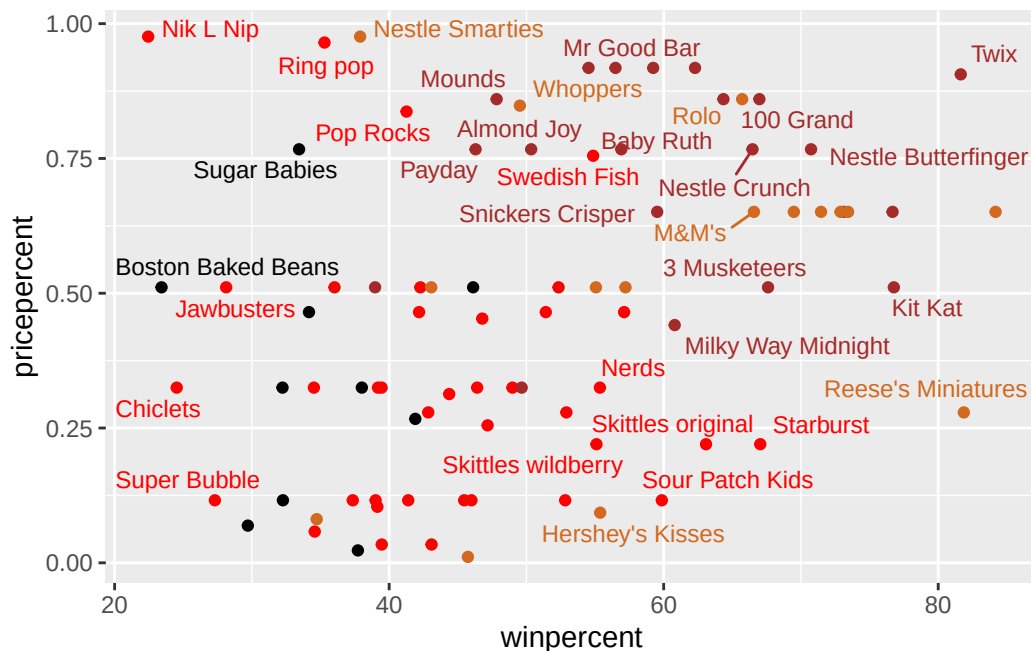
4. Winpercent vs Pricepercent

```
# Pink and gray are too light. lets change to red and black
mycols <- rep("black", nrow(candy))
mycols[as.logical(candy$chocolate)] <- "chocolate"
mycols[as.logical(candy$fruity)] <- "red"
mycols[as.logical(candy$bar)] <- "brown"

library(ggrepel)

# How about a plot of price vs win
ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=mycols) +
  geom_text_repel(col=mycols, size=3.3, max.overlaps = 8)
```

Warning: ggrepel: 52 unlabeled data points (too many overlaps). Consider increasing max.overlaps



. Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

```
ord <- order(candy$pricepercent, decreasing = TRUE)
tail( candy[ord,c(11,12)], n=5 )
```

	pricepercent	winpercent
Strawberry bon bons	0.058	34.57899
Dum Dums	0.034	39.46056
Fruit Chews	0.034	43.08892
Pixie Sticks	0.023	37.72234
Tootsie Roll Midgies	0.011	45.73675

. Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
ord <- order(candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719

Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

5. Correlation Structure

```
cij <- cor(candy)
cij
```

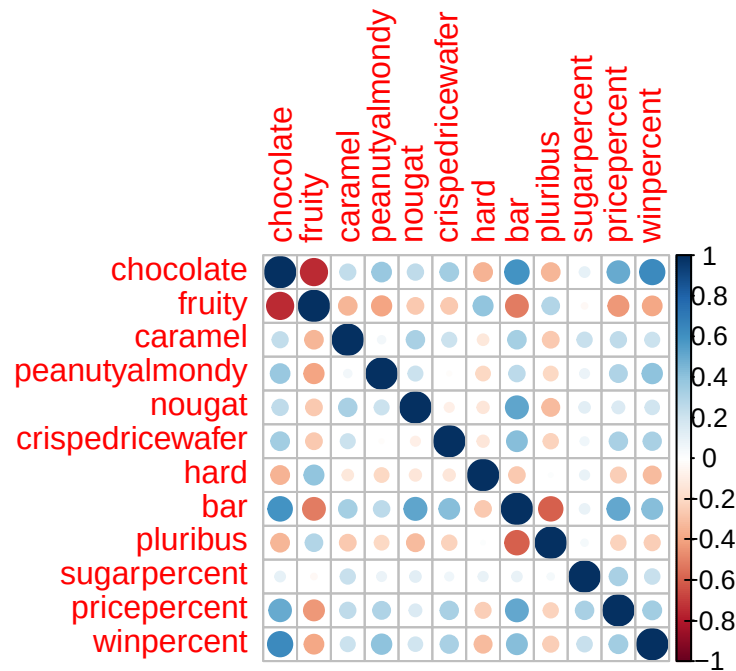
	chocolate	fruity	caramel	peanutyalmondy	nougat
chocolate	1.0000000	-0.74172106	0.24987535	0.37782357	0.25489183
fruity	-0.7417211	1.00000000	-0.33548538	-0.39928014	-0.26936712
caramel	0.2498753	-0.33548538	1.00000000	0.05935614	0.32849280
peanutyalmondy	0.3778236	-0.39928014	0.05935614	1.00000000	0.21311310
nougat	0.2548918	-0.26936712	0.32849280	0.21311310	1.00000000
crispedricewafer	0.3412098	-0.26936712	0.21311310	-0.01764631	-0.08974359
hard	-0.3441769	0.39067750	-0.12235513	-0.20555661	-0.13867505
bar	0.5974211	-0.51506558	0.33396002	0.26041960	0.52297636
pluribus	-0.3396752	0.29972522	-0.26958501	-0.20610932	-0.31033884
sugarpercent	0.1041691	-0.03439296	0.22193335	0.08788927	0.12308135
pricepercent	0.5046754	-0.43096853	0.25432709	0.30915323	0.15319643
winpercent	0.6365167	-0.38093814	0.21341630	0.40619220	0.19937530
	crispedricewafer	hard	bar	pluribus	
chocolate	0.34120978	-0.34417691	0.59742114	-0.33967519	
fruity	-0.26936712	0.39067750	-0.51506558	0.29972522	
caramel	0.21311310	-0.12235513	0.33396002	-0.26958501	
peanutyalmondy	-0.01764631	-0.20555661	0.26041960	-0.20610932	
nougat	-0.08974359	-0.13867505	0.52297636	-0.31033884	
crispedricewafer	1.00000000	-0.13867505	0.42375093	-0.22469338	
hard	-0.13867505	1.00000000	-0.26516504	0.01453172	
bar	0.42375093	-0.26516504	1.00000000	-0.59340892	
pluribus	-0.22469338	0.01453172	-0.59340892	1.00000000	
sugarpercent	0.06994969	0.09180975	0.09998516	0.04552282	
pricepercent	0.32826539	-0.24436534	0.51840654	-0.22079363	
winpercent	0.32467965	-0.31038158	0.42992933	-0.24744787	
	sugarpercent	pricepercent	winpercent		
chocolate	0.10416906	0.5046754	0.6365167		
fruity	-0.03439296	-0.4309685	-0.3809381		
caramel	0.22193335	0.2543271	0.2134163		
peanutyalmondy	0.08788927	0.3091532	0.4061922		

nougat	0.12308135	0.1531964	0.1993753
crispedricewafer	0.06994969	0.3282654	0.3246797
hard	0.09180975	-0.2443653	-0.3103816
bar	0.09998516	0.5184065	0.4299293
pluribus	0.04552282	-0.2207936	-0.2474479
sugarpercent	1.00000000	0.3297064	0.2291507
pricepercent	0.32970639	1.0000000	0.3453254
winpercent	0.22915066	0.3453254	1.0000000

```
library(corrplot)
```

```
corrplot 0.95 loaded
```

```
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Chocolate and fruity are negatively correlated

```
round( cij["chocolate", "fruity"], 2)
```

```
[1] -0.74
```

Q23. Similarly, what two variables are most positively correlated?

6. Principle Component Analysis (PCA)

We need to be sure to scale our input `candy` data before PCA as we have the `winpercent` column on a different scale to all others in the dataset.

```
pca <- prcomp(candy, scale=T)
summary(pca)
```

Importance of components:

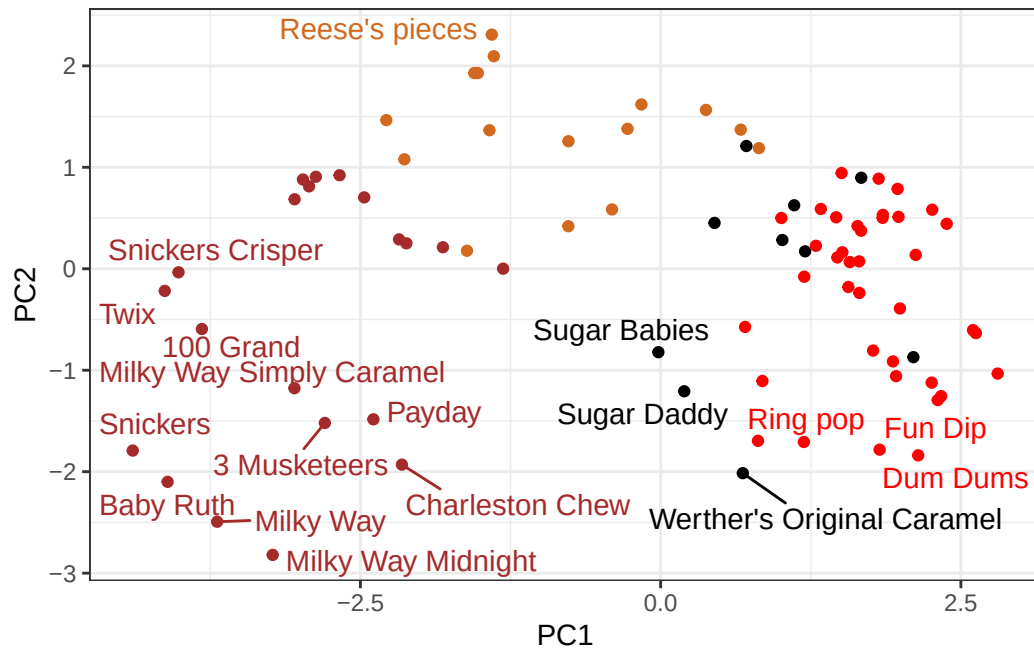
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

First main result figutr is my “PCA plot”

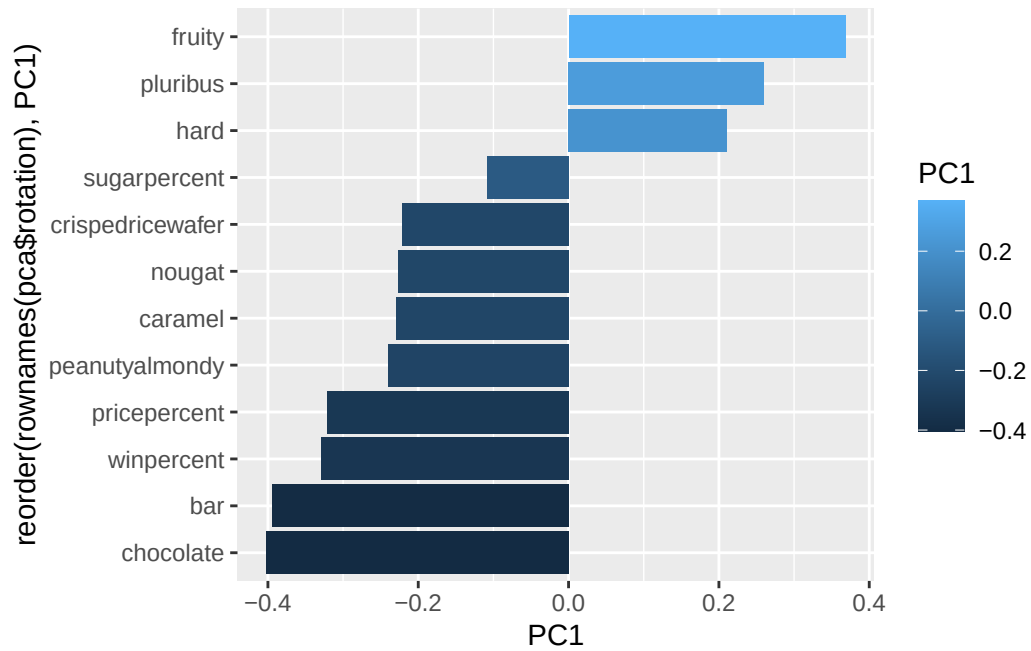
```
#pca$x
ggplot(pca$x) +
  aes(PC1, PC2, label=rownames(pca$x)) +
  geom_point(col=mycols) +
  geom_text_repel(max.overlaps =6, col=mycols) +
  theme_bw()
```

Warning: ggrepel: 67 unlabeled data points (too many overlaps). Consider increasing max.overlaps



The second main PCA result is in the `pca$rotation` we can plot this to generate a so-called “loadings” plot.

```
#pca$rotation
ggplot(pca$rotation) +
  aes(PC1, reorder(rownames(pca$rotation), PC1), fill=PC1) +
  geom_col()
```



. Q24. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you?

Fruity, pluribus, and hard. It makes sense to me because fruity and hard candy are popular.